

QA ENGINEER / SDET • CALGARY, AB

Krishna Puppala

I test by risk, a habit built where defects meant recalls. Six years from FDA-regulated medical device manufacturing to enterprise SaaS: Selenium and Playwright frameworks in Java, Python, and TypeScript, API and MS-SQL data validation, CI/CD quality gates. Currently building a Playwright-TypeScript framework from scratch at Walnut Insurance.

[GitHub](https://github.com/krishnaharshap)(https://github.com/krishnaharshap) [LinkedIn](https://www.linkedin.com/in/krishna-p-472514236/)(https://www.linkedin.com/in/krishna-p-472514236/)

Email

THIS PAGE TESTS ITSELF

RISK content drifts from the source resume

SPEC tests/content.spec.ts

CI Playwright on every push [view runs](#)

SKILLS

Tooling chosen by risk, not fashion

Test Automation

Selenium WebDriver / Playwright (TypeScript, Python, Java) / Cucumber BDD (Gherkin) / TestNG / JUnit / PyTest / Appium / Page Object Model

Languages and Data

Java / Python / TypeScript / JavaScript / MS-SQL (complex queries, CTEs, joins) / Backend Reconciliation / Data Integrity Validation

CI/CD and Tooling

Jenkins / GitHub Actions / Azure DevOps / Git / GitHub / Bitbucket / Maven / Docker / BrowserStack / Testim.io

Test Types

Functional / Regression / Integration / End-to-End / API / Smoke / Performance and Load / Data Integrity / Exception Reporting / UAT

Regulated and Compliance

Medical Device V&V / FDA Design Controls / Risk-Based Test Strategy / Requirement Traceability / CAPA Investigation / ISO QMS Frameworks

API and Performance

Postman / REST Assured / REST API Validation / JMeter / LoadRunner

AI-Assisted Testing

AI-assisted test case generation / LLM-driven automation workflows (Claude, Copilot) / Flaky test triage and root cause analysis

Test Management

Jira / TestRail / Zephyr Squad / qTest / HP ALM / ServiceNow / Confluence

EXPERIENCE • 2023 - PRESENT

Shipping insurance platforms without Sev-1s

Playwright-TypeScript framework, built from zero

QA Engineer (Internship)

June 2026 - Present

Walnut Insurance | Calgary, AB

- Building a Playwright-TypeScript automation framework from scratch, covering UI and API test scenarios for core insurance products
 - Applying AI-assisted test case generation to accelerate coverage for new product features while keeping every generated case traceable to an acceptance criterion
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IT Fundamentals Certificate

Full-time upskilling

Dec 2025 - May 2026

Southern Alberta Institute of Technology (SAIT) | Calgary, AB

Completed the Information Technology Fundamentals Certificate full-time: IT business analysis, project management, data concepts, digital solutioning, Python programming and data analytics, Agile project management

Zero Sev-1 production defects across monthly releases

Senior Quality Assurance Analyst

Dec 2023 - Nov 2025

Sovereign General Insurance Company (SGIC) | Calgary, AB

- Designed and executed 1,000+ manual and automated test cases against user stories and acceptance criteria, raising pre-release defect detection by roughly 30% through Selenium and Cucumber regression suites and cutting regression cycle time by about half
 - Led full testing effort as the sole QA in an Agile squad on a Tomcat-based web service front end, delivering stable monthly releases with zero Sev-1 production defects; critical defects caught pre-release in underwriting workflows were estimated by business operations to have avoided millions annually in underwriter rework
 - Wrote complex MS-SQL queries (CTEs, multi-table joins) to validate backend data accuracy and business rule compliance across policy rating engines and legacy AS400-integrated services, supporting reconciliation checks and defect root cause investigation
 - Performed REST API validation in Postman with scenario-based verification, payload validation, and response time analysis, sustaining about 99% integration reliability across interconnected enterprise applications
 - Integrated automated regression and smoke suites into Jenkins CI pipelines with pre-deployment quality gates, reducing production rollback incidents
 - Led initial test planning for legacy application server migrations from Windows Server 2016 to 2022, including validation of pre-production SQL Server database migrations
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EXPERIENCE • 2017 - 2023

Trained where defects cost recalls

80% of defects caught before UAT

Senior Quality Assurance Analyst

Jan 2023 - Dec 2023

The AppLabb | Clients: Suncor Energy, Rise Health, Frank Mortgage | Calgary, AB

- Performed functional, integration, and API testing across enterprise web and cross-platform mobile applications, catching 80% of defects before UAT across client build cycles
 - Validated UI, API, and backend SQL data flows end to end for AI-enabled applications hosted on Microsoft Azure, including a mortgage broker platform with an AI chat interface
 - Managed the defect lifecycle in Jira and ServiceNow across onshore and offshore teams: logging, triage, retesting, impact analysis, and release validation
 - Automated mobile test execution with Testim.io and BrowserStack cloud devices, introduced Android simulators across OS versions, and maintained JMeter and LoadRunner performance scripts
 - Coordinated UAT cycles and release sign-offs, communicating testing risks and release blockers to product owners and business stakeholders
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Safety-critical devices under FDA design controls

Quality Engineer (Verification & Validation)

Dec 2018 - Sep 2021

Fraank Systems | Clients: Baxter Healthcare, Roche Diagnostics, Mentor Worldwide | United States

- Executed verification, validation, and traceability activities for safety-critical medical devices in compliance with ISO quality standards and regulated QMS frameworks
- Designed test protocols and executed validation cycles under FDA design control requirements, with reliability focus, risk assessment, and failure analysis
- Built requirement traceability matrices and supported CAPA investigation documentation, collaborating with software, hardware, and QA teams on continuous quality improvement

Design verification in regulated manufacturing

Design Quality Engineer

Mar 2017 - Dec 2018

Vastek Group | Clients: Zimmer Biomet | Warsaw, IN, United States

- Performed design verification testing for electromechanical medical devices through a structured V&V lifecycle in a regulated manufacturing environment
 - Documented defect investigations and traceability reports linking requirements to validation outcomes, ensuring compliance with FDA and ISO standards through the product development lifecycle
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PROJECTS

Proof you can run

Cross-browser E2E suite with a live, public test report

selenium-cucumber-framework

Java + Selenium 4 + Cucumber 7 BDD framework for an e-commerce demo app: Page Object Model, parallel cross-browser runs (Chrome, Firefox, Edge), Allure and Extent reporting, screenshot capture on failure, GitHub Actions and Jenkins pipelines.

Java, Selenium, Cucumber, TestNG, Maven, Allure

[Repository](#) [Live test report](#)

Browser automation driven through the Playwright MCP server

playwright-python-mcp

Python client that drives browser automation by talking to the official Playwright MCP server over stdio. Smoke tests against TodoMVC, unit tests plus optional live integration tests, GitHub Actions CI.

Python, Playwright MCP, pytest, Node.js

[Repository](#)

Banking demo UI suite with CI artifacts

ui-automation-parabank

Playwright for Java + TestNG automation for the ParaBank banking app: login flows, form validation, headless smoke tests, Extent and Surefire reporting, GitHub Actions CI with artifact uploads.

Java, Playwright, TestNG, Maven

[Repository](#)

Containerized QA tooling, ~20MB image

docker-qa-demo

Alpine-based Docker image demonstrating a full container lifecycle for QA tooling: build, tag, publish to GHCR, with OCI metadata labels and a container health validation script.

Docker, Alpine, GHCR

[Repository](#)

How this site is tested

This page is its own test subject. The strategy is risk-based: each risk maps to a spec, and CI runs the suite on every push.

Content drifts from the source resume

content.spec asserts rendered sections against resume-data.json, including a banned-character scan

Navigation breaks on a browser I did not open

Playwright runs Chromium, Firefox, and WebKit projects in CI on every push

Contrast or focus regressions ship silently

a11y.spec runs axe-core checks plus keyboard traversal assertions

External links rot after the repo goes quiet

A scheduled links workflow probes every external URL with per-host status policies, kept out of the PR gate so it cannot cry wolf

The print and PDF path degrades

print.spec emulates print media and asserts all six sections render linearly

Credentials and contact

Master of Science in Engineering Management

Missouri State University | 2014 - 2016

Bachelor of Technology in Mechanical Engineering

National Institute of Technology (NIT) Warangal |
2008 - 2012

Information Technology Fundamentals Certificate

Southern Alberta Institute of Technology (SAIT),
Calgary | 2026

CCIS IT Bridging Program, Employment Skills Training

Calgary Catholic Immigration Society | 2026

GitHub Foundations (GH-900)

GitHub | 2025

AWS Certified Cloud Practitioner (CLF-C02)

Amazon Web Services | 2024

ISTQB Certified Tester, Foundation Level (CTFL)

International Software Testing Qualifications Board
| 2024

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